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Violent Conflict and Behavior: A Field Experiment in Burundi

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1 Big picture

- **Question.** Does exposure to violent conflict change people's behavior in economically meaningful domains: prosocial behavior, risk-taking, and patience?
- **Setting.** The paper studies rural Burundi after the 1993–2003 civil war. The authors revisit communities from earlier household and community surveys and run incentivized field experiments in 2009.
- **Core empirical design.** The authors combine detailed community-level and household-level exposure to violence with experimental measures of social, risk, and time preferences.
- **Main result.** Individuals and communities more exposed to violence behave more altruistically toward neighbors, are more risk-seeking, and are less patient.
- **Identification concern.** Violence might be targeted rather than random. The paper therefore tests exogeneity using prewar and pre-conflict characteristics and also uses instrumental variables based on distance to Bujumbura and altitude.
- **Real-world validation.** Experimental behavior is complemented with survey-based outcomes: social capital, crop choices, and farm improvements. The patterns broadly line up with the experimental results.
- **Interpretation.** The results suggest that conflict shocks can alter behavior and perhaps preferences, although the authors are careful that the data cannot cleanly distinguish preference changes from changes in beliefs, social context, or learning.
- **Development implication.** Violent shocks can have long-run consequences beyond physical destruction. If conflict changes saving, risk-taking, or social behavior, post-war recovery cannot be understood only as capital reconstruction.
- **Core takeaway.** Exposure to violence is associated with greater prosociality, greater risk-taking, and higher discount rates in an incentivized post-conflict field setting.

2 Incremental contribution of the paper

2.1 Contribution relative to conflict and development research

- Earlier conflict research documents effects of civil war on income, capital, institutions, health, education, and political participation.
- Voors et al. add a behavioral channel: conflict may change the decision primitives that govern savings, investment, risk-taking, and social cooperation.
- The paper therefore broadens the view of conflict from a shock to physical and institutional capital to a shock that can reshape behavior directly.

Interpretation: The incremental contribution is not just another estimate of the economic cost of war. It identifies behavioral margins through which conflict can affect long-run development.

2.2 Contribution relative to endogenous-preferences literature

- Economics often treats preferences as stable in the short and medium run.
- Psychology suggests that severe shocks can have persistent effects on outlook, patience, risk tolerance, and social orientation.
- The paper brings these ideas into development economics by measuring behavior after a major conflict shock.
- It uses incentivized experiments rather than only attitudinal survey questions.

Interpretation: The paper is an empirical bridge between psychology’s view of trauma-induced behavioral change and economics’ use of stable preference parameters.

2.3 Contribution relative to field experiments in poor and post-conflict settings

- Prior experimental studies measured social preferences, risk preferences, or time preferences in developing-country settings.
- This paper is among the first to connect such incentivized preference measures to a quasi-exogenous history of civil-war violence.
- It measures three domains simultaneously: social orientation, risk, and time.
- It also links the experimental measures to real-life outcomes, such as social capital, cash-crop choice, and farm investment.

Interpretation: The paper’s distinctive contribution is combining post-conflict historical exposure, incentivized preference measurement, and real-world behavioral proxies in the same sample.

2.4 Contribution relative to Blattman and Bellows-Miguel style evidence

- Blattman and Bellows-Miguel show that conflict exposure can increase political and civic engagement.
- Voors et al. provide a behavioral-microfoundation channel: conflict exposure is associated with more prosocial behavior and different risk/time behavior.
- The paper therefore helps explain why victimization may be followed by greater engagement rather than only by social breakdown.

Interpretation: The paper’s increment is to go below observed civic outcomes and measure economic decision behavior directly.

3 Data and measurement

3.1 Burundi conflict setting

- Burundi experienced repeated ethnic violence after independence, including a severe civil war after the 1993 assassination of President Melchior Ndadaye.
- Violence involved rebel groups and the Tutsi-dominated army, but much violence targeted civilians rather than combatants.
- The authors emphasize that violence was often broad, indiscriminate, and difficult to predict using standard household or community characteristics.
- The conflict left over 300,000 dead and displaced around 1.2 million people.

Interpretation: The empirical design relies on the idea that much of the local variation in violence was not chosen based on household preferences or economic behavior.

3.2 Survey sample

- The experiment was conducted in March and April 2009.
- The sample contains roughly 300 household heads from 35 rural communities.
- These communities came from a larger panel of 100 communities previously surveyed in 1998 and 2007.

- Of the 35 communities, 24 experienced violence during 1993–2003 and 11 did not.
- The authors combine experiment data with prior household and community variables from the 1998 and 2007 surveys.

Interpretation: The earlier survey waves allow the paper to test whether violence was correlated with pre-existing observables and to control for baseline characteristics.

3.3 Conflict exposure measures

- The main community conflict measure is the percentage of people killed in attacks during 1993–2003.
- The average community has 1.99 percent of the population killed in attacks, with a standard deviation of 4.09 and a maximum of 15.63 percent.
- The paper also uses an attack dummy for whether the village was attacked during 1993–1998.
- The main household-level measure is an individual victimization index based on death, theft, ambush, forced labor, and torture of household members.
- The household index is standardized to mean zero and standard deviation one.

Interpretation: Using both community and household measures helps distinguish exposure to local violence from individual victimization experience.

3.4 Social preference experiment

- The social experiment is a modified social value orientation game.
- Subjects are anonymously matched with another participant from their community.
- Each subject makes six choices between two own-other payoff combinations.
- The design uses only nonnegative payoffs, so it does not confound altruism with competitive preferences.
- The social orientation score ranges from 0 to 100: 0 is fully selfish, 100 maximizes the partner’s payoff, and 50 corresponds to the social optimum.
- The average social preference score is 27.32, indicating generally individualistic choices but with meaningful heterogeneity.

Interpretation: The social experiment is close to a dictator allocation task but with multiple allocation tradeoffs.

3.5 Risk preference experiment

- Risk preferences are measured with a game based on choices between a certain amount and a simple gamble.
- In gain choices, the gamble pays 2,000 FBU with probability 0.3 and zero otherwise.
- In loss choices, the gamble involves losing 2,000 FBU with probability 0.3 and losing zero otherwise.
- The certain amount varies across cards.
- The point at which the subject switches from risky to safe choices identifies the degree of risk aversion.
- The resulting scale runs from 0, risk averse, to 3, risk loving.

Interpretation: The key behavioral result is that violence exposure predicts more risk-seeking choices in the gain domain.

3.6 Time preference experiment

- Subjects choose between receiving 1,000 FBU the next day and receiving a larger amount after two weeks plus one day.
- The delayed amount is $1,000(1 + d)$ FBU, with d taking values from 0 to 1.
- The smallest d at which the subject switches to the delayed payment identifies the discount rate.
- Earlier switching means greater patience; requiring a higher d means greater impatience.
- The average discount rate is about 40 percent, with substantial dispersion.

Interpretation: Violence exposure predicts higher discount rates, which the paper interprets as lower patience.

4 Empirical specifications and identification

4.1 Baseline OLS specification

For individual i in community j , the baseline experimental regression can be written as

$$y_{ij} = \alpha + \beta \text{Conflict}_j + X'_{ij}\gamma + Z'_j\delta + FE + \varepsilon_{ij}, \quad (1)$$

where y_{ij} is a social, risk, or time preference measure, Conflict_j is community exposure to violence, X_{ij} contains household-level controls, and Z_j contains community controls.

- Some columns use only community-level averages.
- Other columns use household-level observations.
- Controls include age, gender, literacy, land holdings, community land inequality, distance to market, conflict over land, ethnic homogeneity, socioeconomic homogeneity, population density, and per capita expenditures.
- Stratum, village, or province fixed effects are added in some columns.

Interpretation: The coefficient β measures whether greater exposure to violence predicts different behavior in the experiment.

4.2 Exogeneity and targeting tests

The exogeneity checks regress exposure to violence and attrition on pre-existing household and community characteristics:

$$\text{Conflict}_j = \alpha + W'_j\theta + u_j. \quad (2)$$

- Table 2 tests whether 1998 and 1993 variables predict later violence.
- It also tests whether the presence of households in the 2009 experiment is predicted by prior characteristics or violence.
- Most pre-conflict and pre-survey characteristics do not predict exposure, supporting the claim that violence was not strongly targeted on observables.

Interpretation: These checks are central because the causal interpretation relies on violence exposure being plausibly unrelated to unobserved determinants of experimental behavior.

4.3 Instrumental variables specification

The paper also estimates two-stage least squares models using distance to Bujumbura and altitude as instruments for conflict exposure:

$$Conflict_j = \pi_0 + \pi_1 \log DistanceToBujumbura_j + \pi_2 \log Altitude_j + X'_j \pi + v_j, \quad (3)$$

$$y_{ij} = \alpha + \beta \widehat{Conflict}_j + X'_{ij} \gamma + Z'_j \delta + FE + \varepsilon_{ij}. \quad (4)$$

- Violence was more intense near Bujumbura and varied with geography.
- The exclusion restriction is that distance and altitude affect preferences only through conflict exposure.
- The authors acknowledge this restriction is contestable but use it as a robustness check.

Interpretation: The IV estimates tend to be larger than OLS and generally reinforce the main direction of the findings.

4.4 Real-world behavior validation

The paper repeats the logic using survey outcomes:

- social capital index as a proxy for social preferences;
- share of cash crops as a proxy for risk-taking;
- expenditures on farm improvements as a proxy for time preference and future orientation.

Interpretation: These outcomes test whether experimental play translates into non-experimental behavior.

5 Incremental interpretation and limitations

5.1 Preference change versus behavior change

- The authors often frame the results as evidence that violence changes preferences.
- They are careful, however, that experiments measure behavior, not preferences directly.
- Behavior can also change because of altered beliefs about others, changes in social norms, learning about one's own attitudes, or changes in the local context.

Interpretation: The safest conclusion is that conflict changes behavior in ways consistent with changes in social, risk, and time preferences.

5.2 Why the signs matter

- More prosocial behavior can help rebuild community cooperation after conflict.
- More risk-taking can raise entrepreneurship and crop diversification, but can also increase exposure to bad shocks.
- Greater impatience can lower investment and savings, potentially slowing long-run recovery.

Interpretation: Conflict does not have a single positive or negative behavioral effect. It changes multiple behavioral margins with potentially offsetting development consequences.

6 Tables and figures

6.1 Table 1: Descriptives

Read:

- Panel A reports the experimental outcomes. The mean social preference score is 27.32 on a 0–100 scale; risk gains average 1.87; risk losses average 2.31; the mean discount rate is 40.16.
- Panel B reports conflict exposure. Average deaths in attacks are 1.99 percent of community population, with substantial variation.
- Panels C and D summarize household and community covariates used in the regressions.

Economic message: The table shows both meaningful behavioral variation and meaningful conflict variation, which make the empirical tests possible.

TABLE 1—DESCRIPTIVES

	Observations household level	Observations community level	Mean	SD	Min	Max
<i>Panel A. Preferences</i>						
Social preferences (0–100, 2009)	286	35	27.32	27.22	0	100
Risk preferences Gains (2009)	220	35	1.87	1.31	0	3
Risk preferences Losses (2009)	233	35	2.31	1.18	0	3
Discount rate (percent, 2009)	273	35	40.16	41.43	0	100
<i>Panel B. Conflict variables</i>						
Relative number of dead in attacks (percent, 1993–2003) ⁽³⁾		35	1.99	4.09	0	15.63
Attack (1993–1998) ⁽⁸⁾		35	0.69	0.47	0	1
Individual victimization index (1993–2003) ⁽²⁾	287	35	0	1	−0.78	5.03
Physical attack (1993–2003) ⁽²⁾	287	35	0.32	0.47	0	1
Nonphysical attack (1993–2003) ⁽²⁾	287	35	0.45	0.50	0	1
<i>Panel C. Household variables</i>						
Household head is literate (1998) ⁽¹⁾	285	35	0.40	0.49	0	1
Household head is literate (2009) ⁽⁴⁾	287	35	1.60	0.49	1	2
Household head age (1998) ⁽¹⁾	283	35	42.28	15.30	16	99
Household head age (2009) ⁽⁴⁾	286	35	45.96	15.11	18	90
Household head is male (1998) ⁽¹⁾	286	35	0.62	0.49	0	1
Household head is male (2009) ⁽⁴⁾	287	35	0.79	0.41	0	1
Household head ethnic origin (2009) ⁽⁴⁾	287	35	0.22	0.41	0	1
Total expenditures (1998) ⁽¹⁾	287	35	8.56	0.70	5.17	10.7
Total land holdings per capita (ha ² , 2007) ⁽²⁾	281	35	5.54	5.13	0.10	34.25
Household head is livestock farmer (1993) ⁽⁴⁾	287	35	0.41	0.49	0	1
Perceived trust level (1998) ⁽²⁾	287	35	4.64	2.17	1	10
Social capital index (2007) ⁽²⁾	282	35	0.02	1.04	−0.47	4.49
Investments farm buildings (FBU, 1998) ⁽¹⁾	241	35	5.57	1.20	1.08	8.52
Investments farm buildings (FBU, 2007) ⁽²⁾	287	35	229.35	1,022.67	0	12,155.87
Share of cash crops in total production (1998) ⁽¹⁾	276	35	0.22	0.30	0	1
Share of cash crops in total production (2007) ⁽²⁾	280	35	0.06	0.14	0	1
Severe draught (percent yes, 2007–2009) ⁽⁴⁾	277	35	0.33	0.47	0	1
Access rain (percent yes 2007–2009) ⁽⁴⁾	231	35	0.62	0.49	0	1
Manioc disease (percent yes 2007–2009) ⁽⁹⁾	287	35	0.40	0.49	0	1
Upcoming ceremony (percent yes, 2009) ⁽⁴⁾	231	35	0.39	0.49	0	1
<i>Panel D. Community variables</i>						
Land Gini coefficient (2007) ⁽²⁾	35	0.29	0.20	0	0.54	
Distance to market (2007) ⁽³⁾	35	2.86	0.72	1.38	4.22	
Conflict over land (percent yes, 2007) ⁽²⁾	35	0.25	0.15	0	0.6	
Ethnic homogeneity (1993) ⁽⁵⁾	31	82.87	16.89	30	99	
Ethnic homogeneity (2009) ⁽⁵⁾	35	86.68	15.57	30	100	
Votes for Ndadaye in 1993 (percent) ⁽⁷⁾	34	64.51	17.68	9.94	93.51	
Socioeconomic homogeneity (1998) ⁽²⁾	35	1.66	1.06	1	5	
Socioeconomic homogeneity (2007) ⁽²⁾	35	1.57	0.95	1	5	
Population density (log, 1990) ⁽⁵⁾	35	5.43	0.51	4.20	6.11	
Population density (log, 2008) ⁽⁶⁾	35	5.76	0.47	4.50	6.49	
Per capita total expenditure (log, 2007) ⁽²⁾	35	9.25	0.47	8.09	10.40	
Distance to Bujumbura (km, log)	35	4.49	0.39	3.68	5.12	
Altitude (m, log) ⁽⁵⁾	35	7.41	0.10	7.14	7.70	

Sources: (1) Burundi Priority Household Survey 1998, (2) Burundi Priority Household Survey 2007, (3) Burundi Community Survey 2007, (4) Burundi Experiments Exit Survey 2009, (5) Monographies Communales Burundi. Ministère de la Planification du Développement et de la Reconstruction Nationale, Bujumbura, 2006 (6) Recensement National de la Population et de l'Habitat 2008, (7) Sinunguruzi T. 2001. Les Elections au Burundi. Tout Savoir et Tirer des Leçons de Juin. (8) Raleigh et al. 2010. (9) FAO Valuation Des Recoltes. Des Approvisionnements

6.2 Table 2: Exogeneity and attrition

Read:

- Table 2 tests whether pre-existing characteristics predict conflict exposure and sample attrition.
- Most household and community observables do not predict violence or later survey presence.
- Geographic variables, especially distance to Bujumbura and altitude, are strongly related to conflict exposure and are later used as instruments.

Economic message: The table supports the claim that observed patterns are not driven by obvious targeting of violence to pre-conflict socioeconomic characteristics.

TABLE 2—EXOGENEITY

Dependent variable	Attack during 1993–2003	Percentage dead in attacks 1993–2003	Percentage dead in attacks 1993–2003	Physical attack on household member	Nonphysical attack on household member	Present in village in 1998 and 2009	Present in village in 1993 and 1998
	Probit (1)	OLS (2)	OLS (3)	Probit (4)	Probit (5)	Probit (6)	Probit (7)
Percentage literate household heads (in 1998)	–0.0282 [0.0208]	–0.0561 [0.041]	–0.002 [0.001]				
Average age household head (in 1998)	–0.0273 [0.0520]	0.0625 [0.133]	–0.001 [0.005]				
Percent male (in 1998)	–0.00170 [0.0248]	–0.0587 [0.0634]	0.003 [0.002]				
Percentage livestock farmers (in 1993)	0.0252 [0.0218]	0.0524 [0.0510]					
Density in 1990 (log)	–2.119 [1.443]	0.0317 [1.970]	0.081 [0.123]				
Ethnic homogeneity (in 1993)	0.0155 [0.0235]	0.0200 [0.0557]					
Socioeconomic homogeneity (in 1998)	–0.295 [0.392]	–0.934 [0.872]	0.051 [0.033]				
Percentage of votes for Ndayaye (in 1993)	0.0227 [0.0283]	0.0457 [0.0575]					
Distance to Bujumbura (km, log)	–2.561 [1.536]*	–4.884 [3.039]*	0.109 [0.189]				
Altitude (m, log)	–8.112 [5.420]*	–16.09 [8.252]*	–0.748 [0.303]**				
Relative number of dead in attacks 1993–2003							0.0255 [0.0345]
Number of attacks 1993–1998							–0.001 [0.002]

(Continued)

Table 2, part 1

TABLE 2—EXOGENEITY (Continued)

Dependent variable	Attack during 1993–2003	Percentage dead in attacks 1993–2003	Percentage dead in attacks 1993–2003	Physical attack on household member	Nonphysical attack on household member	Present in village in 1998 and 2009	Present in village in 1993 and 1998
	Probit (1)	OLS (2)	OLS (3)	Probit (4)	Probit (5)	Probit (6)	Probit (7)
Respondent is literate (in 1998)				–0.114 [0.251]	0.224 [0.234]	0.312 [0.208]	0.023 [0.137]
Respondent age (in 1998)				–0.005 [0.008]	0.006 [0.008]	–0.002 [0.006]	0.012 [0.003]***
Respondent is male (in 1998)				0.447 [0.303]	–0.248 [0.278]	–0.794 [0.225]***	–0.054 [0.083]
Livestock farmer in 1993				0.228 [0.230]	0.0958 [0.214]	0.0660 [0.226]	–0.0003 [0.084]
Respondent ethnicity				–0.498 [0.310]	–0.473 [0.296]		
Total expenditures (in 1998)				0.361 [0.222]	0.345 [0.217]	–0.193 [0.170]	
Perceived trust level (in 1998)				–0.001 [0.051]	0.028 [0.046]		
Share of cash crops in total production (in 1998)				0.176 [0.484]	–0.499 [0.434]	–0.097 [0.367]	
Expenditures farm improvement (in 1998)				0.0949 [0.171]	0.237 [0.161]	0.030 [0.098]	
Constant	83.37 [52.05]	142.5 [74.01]*	4.784 [2.473]*	–10.06 [0.00]	–4.199 [1.951]*	2.295 [1.448]	–0.993 [0.215]***
FE	No	No	Yes (province level)	Yes (village level)	Yes (village level)	Yes (stratum level)	Yes (province level)
N	30	30	94	198	219	279	1,766
Adjusted R ²	0.33	0.19	0.27	0.19	0.16	0.10	0.11

Notes: Included fixed effects are at lowest level possible. Columns 1 and 2 use 30 observations instead of 35 due to missing 1993 ethnicity data. Column 3 uses 94 observations and not 100 due to missing 1993 population data. Dependent variable dead in attacks relative to number of households and not population due to missing population data. Columns 1–6 use mainly BPHS and BSC 2009 data. Ethnic livestock ownership data are from BIEES (2009). Dependent variable in column 6 is a dummy, 1 if respondent was interviewed in both 1998 and 2007, zero else. Literacy is measured as years of education. Column 7 uses the ESD-SR 2002 data to assess attrition between 1993 and 1998. Dependent variable is a dummy, 1 if respondent was present in village in 1993 and 1998, zero else. Literacy, age, and gender are measured in 2002; the dummy for livestock farmer in 1993 is based on recall in 2002, the number of attacks between 1993 and 1998 were drawn from the ACLED database and matched to the UNFPA data at the commune level. The proportion of people absent from the sample was 16 percent. Regression uses province fixed effects. Including village fixed effects reduces the number of observations but does not change the results. Standard errors in brackets.

*** Significant at the 1 percent level.

Table 2, part 2

6.3 Table 3: Conflict and social preferences

Read:

- The dependent variable is the social orientation score.
- The percentage dead in attacks is positively associated with altruistic choices across OLS specifications.
- The individual victimization index is also positively associated with altruistic behavior.
- 2SLS estimates using distance to Bujumbura and altitude are positive and significant.

Economic message: Exposure to conflict predicts greater prosocial behavior toward neighbors.

6.4 Table 4: Conflict and risk preferences

Read:

- The dependent variable ranges from 0, risk averse, to 3, risk loving.

- Violence exposure predicts more risk-seeking behavior in the gains domain.
- The coefficient is positive across OLS and IV specifications.
- The loss-domain result is weaker.

Economic message: Conflict exposure is associated with greater willingness to take risks, especially over gains.

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	2SLS (6)	2SLS (7)
Percentage dead in attacks	1.073 [0.489]**	0.875 [0.460]*	1.688 [0.464]***	1.686 [0.523]***		2.892 [0.958]***	0.486
Individual victimization index					2.940 [1.745]*		
Respondent is literate			9.550 [3.521]***	9.430 [3.450]***	6.645 [3.945]*	10.39 [3.418]***	0.190
Respondent age			-0.207 [0.108]*	-0.201 [0.109]*	-0.330 [0.158]**	-0.202 [0.110]*	-0.111
Respondent is male			4.204 [3.373]	4.662 [3.436]	10.97 [4.234]**	5.615 [3.174]*	0.099
Total land holdings per capita			0.807 [0.253]***	0.730 [0.250]***	0.590 [0.380]	0.827 [0.263]***	0.152
Land Gini coefficient			-16.47 [10.01]	-13.60 [9.376]		-20.66 [12.12]*	-0.150
Distance to market			-6.012 [2.583]**	-6.020 [2.542]**		-7.633 [2.851]***	-0.198
Conflict over land			-29.99 [11.65]**	-26.79 [10.74]**		-36.77 [13.32]***	-0.207
Ethnic homogeneity			0.204 [0.118]*	0.232 [0.124]*		0.148 [0.126]	0.080
Socioeconomic homogeneity			-2.387 [1.948]	-2.245 [2.154]		-2.244 [2.077]	-0.072
Population density			6.712 [4.173]	5.770 [5.395]		8.511 [6.074]	0.136*
Per capita total expenditure			1.995 [3.046]	2.610 [3.845]		0.947 [4.483]	0.016
Constant	23.688 [2.569]***	25.20 [2.296]***	-23.68 [42.45]	-24.86 [42.63]	4.33 [34.16]	-12.17 [44.99]	
FE	No	No	No	Yes (stratum level)	Yes (village level)	Yes (stratum level)	Yes (stratum level)
1998 household controls	No	No	No	No	Yes	No	No
N	35	286	278	278	225	278	278
Adjusted R ²	0.099	0.018	0.144	0.140	0.212	0.166	0.166

First stage instruments

Distance to Bujumbura (log)	-6.687 [2.489]***
Altitude (log)	-21.988 [9.211]***
Hansen J, p-value	0.51
Partial F	7.30

Notes: Dependent variable: degree of altruism scale 0–100. Robust standard errors in brackets are clustered at community level. Sampling weights are not included, but results are qualitatively identical. Column 1: dependent variable is community average. Column 5: household 1998 controls included. Column 6: excluded instruments of first stage reported only. The beta coefficients of column 6 are reported in column 7. Estimations in column 7 do not

	Gains OLS (1)	Gains OLS (2)	Gains OLS (3)	Gains OLS (4)	Gains OLS (5)	Losses OLS (6)	Gains 2SLS (7)	Gains 2SLS (8)
Percentage dead in attacks	0.078 [0.024]***	0.0651 [0.0246]**	0.0527 [0.0246]**	0.0634 [0.0263]**		0.0196 [0.0204]	0.0729 [0.0376]*	0.258
Individual victimization index					0.165 [0.114]*			
Respondent is literate			-0.225 [0.179]	-0.227 [0.183]	-0.005 [0.274]	-0.151 [0.182]	-0.214 [0.170]	-0.083
Respondent age			0.005 [0.006]	0.006 [0.006]	0.001 [0.009]	-0.008 [0.006]	0.006 [0.005]	0.065
Respondent is male			-0.217 [0.188]	-0.186 [0.188]	-0.279 [0.256]	0.0537 [0.141]	-0.176 [0.165]	-0.067
Total land holdings per capita			-0.025 [0.018]	-0.0341 [0.018]*	-0.028 [0.025]	-0.010 [0.019]	-0.034 [0.017]*	-0.130
Land Gini coefficient			-1.129 [0.745]	-1.026 [0.725]		0.0393 [0.731]	-1.092 [0.740]	-0.170
Distance to market			0.124 [0.129]	0.137 [0.122]		0.125 [0.141]	0.125 [0.129]	0.070
Conflict over land			0.0205 [0.960]	0.254 [0.918]		0.510 [0.877]	0.167 [1.045]	0.019
Ethnic homogeneity			-0.000878 [0.00641]	-0.0000909 [0.00671]		0.00161 [0.00877]	-0.001 [0.006]	0.009
Socioeconomic homogeneity			-0.00605 [0.115]	0.0314 [0.117]		-0.0381 [0.136]	0.0313 [0.111]	0.022
Population density			0.326 [0.225]	0.418 [0.332]		0.430 [0.292]	0.441 [0.333]	0.149
Per capita total expenditure			-0.249 [0.143]*	-0.274 [0.170]		-0.0302 [0.239]	-0.290 [0.177]	-0.106
Constant	1.680 [0.131]***	1.716 [0.145]***	2.237 [1.986]	2.022 [2.116]	4.902 [1.869]**	0.00443 [2.607]	2.149 [2.045]	
FE	No	No	No	Yes (stratum level)	Yes (village level)	Yes (stratum level)	Yes (stratum level)	Yes (stratum level)
1998 household controls	No	No	No	No	Yes	No	No	No
N	35	220	213	213	175	228	213	213
Adjusted R ²	0.17	0.05	0.18	0.20	0.38	0.09	0.20	0.20

First stage instruments

Distance to Bujumbura (log)	-5.499 [2.525]*
Altitude (log)	-19.221 [8.474]**
Hansen J, p-value	0.10
Partial F	6.50

Notes: Dependent variable ranges from 0 (risk averse) to 3 (risk loving). Robust standard errors in brackets are clustered at community level. Sampling weights are not included, and results are qualitatively identical. Column 1: dependent variable is community average. Column 5: household 1998 controls included. Column 7: excluded

6.5 Table 5: Conflict and time preferences

Read:

- The dependent variable is the elicited discount rate.
- Higher conflict exposure is associated with higher discount rates in several specifications.
- The IV estimates are also positive and significant.
- Individual victimization has a positive but imprecise coefficient.

Economic message: Conflict exposure predicts greater impatience, which can have implications for savings and investment.

6.6 Table 6: Real-world behavior and decisions

Read:

- The paper replaces experimental outcomes with survey-based proxies.

TABLE 5—CONFLICT AND TIME PREFERENCES

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	2SLS (6)	2SLS (7)
Percentage dead in attacks	0.543 [0.519]	0.666 [0.467]	1.197 [0.624]*	1.188 [0.575]**		2.337 [1.058]**	0.265
Individual victimization index					2.826 [3.294]		
Respondent literate			-5.540 [5.594]	-5.051 [5.610]	-12.51 [6.463]*	-4.136 [5.405]	-0.050
Respondent age			-0.162 [0.189]	-0.174 [0.186]	-0.254 [0.267]	-0.180 [0.178]	-0.065
Respondent is male			-0.168 [4.921]	-1.225 [4.747]	-11.42 [7.483]	-0.382 [4.470]	-0.004
Total land holdings per capita			0.154 [0.442]	0.331 [0.439]	-0.516 [0.633]	0.423 [0.408]	0.053
Land Gini coefficient			-32.65 [23.40]	-39.88 [22.50]*		-47.18 [21.50]**	-0.230
Distance to market			3.013 [3.704]	2.873 [4.211]		1.318 [4.769]	0.023
Conflict over land			-14.49 [24.33]	-22.92 [22.57]		-33.35 [22.80]	-0.124
Ethnic homogeneity			-0.432 [0.224]*	-0.496 [0.212]**		-0.582 [0.195]**	-0.207
Socioeconomic homogeneity			7.287 [4.516]	6.839 [4.928]		6.711 [4.881]	0.144
Population density			7.489 [6.715]	9.804 [7.421]		12.48 [7.999]	0.132
Per capita total expenditure			8.874 [6.479]	7.632 [5.267]		5.825 [5.564]	0.066
Constant	39.38 [4.377]***	38.49 [3.950]***		-44.73 [72.99]	-45.15 [70.65]	74.50 [64.10]	-29.89 [71.18]
FE	No	No	No	Yes (stratum level)	Yes (village level)	Yes (stratum level)	Yes (stratum level)
1998 household controls	No	No	No	No	Yes	No	No
N	35	273	266	266	213	266	266
Adjusted R ²	0.01	0.01	0.15	0.16	0.40	0.15	0.15
<i>First stage instruments</i>							
Distance to Bujumbura (log)						-6.605 [2.519]**	
Altitude (log)						-22.488 [9.388]**	
Hansen J, <i>p</i> -value						0.97	
Partial F						7.25	

Notes: Dependent variable: discount rate. Robust standard errors in brackets clustered at community level. Column 1: dependent variable is community average. Sampling weights are not included, and results are qualitatively identical. Column 5: household 1998 controls included. Column 6: excluded instruments of first stage reported only. Column 7 contains beta coefficients of column 6. Estimations in column 7 do not include clustered standard errors.

TABLE 6—CONFLICT AND REAL WORLD BEHAVIOR AND DECISIONS

Dependent variable	Social capital	Share of cash crops in total production	Expenditures on farm improvements	Social sub-sample 1998 same gender households	Risk sub-sample 1998 same gender households	Time sub-sample 1998 same gender households
	OLS (1)	Tobit (2)	OLS (3)	2SLS (4)	2SLS (5)	2SLS (6)
Percentage dead in attacks	0.033 [0.016]**	0.008 [0.005]*	-23.75 [12.810]*	3.080 [1.294]**	0.0930 [0.0574]*	0.424 [1.246]
Respondent is literate	0.374 [0.150]**	0.041 [0.039]	113.8 [92.48]	12.30 [4.625]**	-0.0957 [0.190]	-9.317 [5.915]
Respondent age	-0.001 [0.004]	0.002 [0.001]	-1.019 [3.598]	-0.346 [0.160]**	0.00154 [0.00657]	-0.249 [0.212]
Respondent is male	-0.111 [0.153]	0.079 [0.039]**	-35.47 [78.75]	-1.625 [5.255]	-0.267 [0.215]	7.504 [6.984]
Total land holdings per capita	0.027 [0.015]*	0.011 [0.004]***	-7.243 [9.346]	0.935 [0.309]**	-0.0304 [0.0197]	0.158 [0.428]
Land Gini coefficient	0.097 [0.353]	-0.465 [0.165]**	261.6 [291.7]	-16.85 [16.67]	-0.871 [0.996]	-57.04 [21.62]***
Distance to market	-0.092 [0.107]	-0.003 [0.043]	79.32 [54.88]	-8.053 [3.520]**	0.0952 [0.167]	6.209 [4.456]
Conflict over land	-0.897 [0.445]*	-0.530 [0.180]**	656.0 [464.9]	-33.90 [19.73]*	0.149 [1.318]	-40.05 [23.88]*
Ethnic homogeneity	0.003 [0.007]	0.002 [0.002]	-4.985 [3.597]	0.165 [0.148]	0.000822 [0.00725]	-0.599 [0.210]**
Socioeconomic homogeneity	0.044 [0.124]	0.011 [0.038]	39.08 [53.49]	-1.354 [2.698]	0.130 [0.158]	7.491 [5.159]
Population density	0.014 [0.186]	0.304 [0.103]**	-62.60 [159.5]	8.192 [6.769]	0.536 [0.374]	15.22 [8.582]*
Per capita total expenditure	-0.271 [0.139]*	0.099 [0.068]	113.5 [78.08]	1.172 [6.197]	-0.331 [0.254]	5.042 [5.455]
Constant	2.180 [1.863]	-3.094 [0.670]**	-635.3 [1261.1]	-3.600 [58.28]	1.793 [2.643]	-45.52 [74.99]
FE	Yes	Yes	Yes	Yes	Yes	Yes
N	274	277	279	212	158	205
Adjusted R ²		0.09	0.06	0.14	0.18	0.23
Hansen J, <i>p</i> -value				0.60	0.14	0.75
Partial F				6.09	5.07	5.99

Notes: Robust standard errors in brackets are clustered at community level. Sampling weights are not included, and results are qualitatively identical. Column 2 contains an unconditional Tobit regression with fixed effects, potentially inducing some bias in our estimates compared to using random effects in multilevel models (columns 4, 5, 6). First

- Conflict exposure predicts higher social capital.
- It also predicts a greater share of cash crops in production, consistent with more risk-taking.
- The relationship with farm improvement expenditures is negative in one specification, consistent with lower patience.

Economic message: The experimental findings are reflected in real-world behavior, strengthening the interpretation of the experiments.

6.7 Table 7: Preferences and nonviolent events

Read:

- Table 7 checks whether nonviolent shocks explain behavior.
- Severe drought predicts social preferences, but the other nonviolent events show weaker and less systematic patterns.
- Manioc disease and rain access do not explain the risk measure in the same way as conflict.

Economic message: Violent conflict appears distinctive. The behavioral patterns are not simply responses to any adverse shock.

TABLE 7—PREFERENCES AND NONVIOLENT EVENTS

	OLS (1)	OLS (2)	OLS (3)
Dependent variable	Social	Risk	Time
Severe draught	11.644 [2.965]***	−0.019 [0.204]	2.905 [4.160]
Excess rain	−1.674 [4.325]	−0.039 [0.213]	−1.183 [7.214]
Manioc crop disease	9.451 [6.464]	0.156 [0.265]	5.653 [7.145]
Upcoming ceremony			2.894 [5.967]

Notes: Table summarizes coefficients of separate regressions including same controls and fixed effects as used in Tables 3–5. Robust standard errors in brackets, clustered at community level. Sampling weights are not included, and results are qualitatively identical.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

7 Short summary

- The paper studies rural Burundi after civil war using incentivized field experiments.
- It measures social preferences, risk preferences, and time preferences among 300 household heads in 35 communities.
- Conflict exposure is measured at the community level and household victimization level.
- Exogeneity tests show limited evidence that violence was targeted based on pre-existing socioeconomic characteristics.
- More exposed individuals behave more altruistically, more risk-seekingly, and less patiently.
- IV estimates using distance to Bujumbura and altitude generally reinforce the OLS results.
- Survey-based outcomes such as social capital, cash-crop choice, and farm improvements support the experimental patterns.
- The paper contributes evidence that violent conflict can reshape economic behavior, possibly through changes in preferences.

8 Conclusion

8.1 Core conclusion

Voors et al. show that exposure to violent conflict in Burundi is associated with systematic changes in behavior. Individuals exposed to conflict behave more prosocially toward neighbors, make riskier choices, and display higher discount rates.

8.2 Economic intuition

Violence can change how people perceive others, risk, and the future. The same shock can create stronger local solidarity, greater willingness to gamble, and more impatience. These behavioral responses may help explain why conflict has long-run effects beyond destroyed assets and disrupted institutions.

8.3 Takeaway

The paper’s main lesson is that post-conflict recovery depends not only on rebuilding physical capital. Conflict can alter economic behavior itself. Development models and policy responses should therefore consider social,

risk, and time preference channels after violence.